SURYA S

SOFTWARE ENGINEER & FULL-STACK DEVELOPER | B.E. COMPUTER SCIENCE & ENGINEERING

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PROFESSIONAL SUMMARY

Third-year **Computer Science and Engineering student at Dhanalakshmi Srinivasan College of Engineering, Coimbatore** passionate about software engineering, full-stack web development, and practical digital products. Demonstrated experience building live production platforms (**PosterKadai**, **KisanConnect**), hackathon solutions (**TN Scheme AI at Smartathon '26**), and sensor-driven applications (**Smart Fertilization System at Barnstromz 2K26**). Strong foundation in Object-Oriented Programming, Data Structures, REST APIs, and database engineering across Java, Python, JavaScript, TypeScript, React, Node.js, and MongoDB. Eager to contribute technical competence to software development internships, campus placements, and engineering teams.

TECHNICAL SKILLS

Programming Languages:	Java (Core OOP, Collections), Python, C, C++, JavaScript (ES6+), TypeScript, SQL.
Frontend Development:	React 18, Vite, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive UI/UX Design, DOM Manipulation.
Backend Development:	Node.js, Express.js, FastAPI, RESTful Web Services, JSON APIs, Authentication, Middleware.
Databases & Cloud:	MongoDB Atlas (Atomic Operators, Aggregations), Firebase (Firestore, Hosting), SQLite, MySQL.
Developer Tools & Hardware:	Git, GitHub, Postman, Linux CLI, VS Code, ESP32 Microcontrollers, Arduino, IoT Sensors.
Core CS Foundations:	Data Structures & Algorithms, Object-Oriented System Architecture, Clean Code Principles, SDLC.

FEATURED SOFTWARE ENGINEERING PROJECTS

PosterKadai — E-Commerce Storefront & Personalized Print Business **Live Web App, 2025 – Present**

Technologies: React 18, Firebase Cloud Firestore, Tailwind CSS, WhatsApp Business API

- Architected and shipped an active production commercial storefront (posterkadai-e364b.web.app) for customized posters and wall artwork.
- Implemented responsive client-side shopping cart management with instant real-time pricing and dynamic category browsing.
- Integrated automated WhatsApp checkout bridge delivering structured cart payloads directly to order fulfillment, achieving sub-1.2s page load speeds.

KisanConnect — AgriTech Direct Marketplace Platform **Live Deployment, 2024 – Present**

Technologies: React 18, Node.js, Express.js, MongoDB Atlas, Firebase Hosting

- Developed a full-stack agricultural marketplace platform eliminating intermediary markups, created as part of the MongoDB Tech-Odyssey initiative.
- Implemented real-time APMC Mandi market price comparison indicators (UP, DOWN, STABLE) allowing growers to benchmark transparent produce valuations.
- Designed race-condition-free inventory decrementing utilizing MongoDB atomic operators ($gte, $inc) across Farmer, Consumer, Retailer, and Admin tiers.

TN Scheme AI — Smartathon '26 Government Welfare Discovery Platform **Smartathon '26 Hackathon, 2026**

Technologies: FastAPI, Python, React 18, MongoDB Atlas, Tailwind CSS

- Engineered an automated scheme discovery system during the DSU × ONEYES Smartathon '26 (9–10 September 2026, Problem Statement #6).
- Implemented deterministic rule-matching algorithms evaluating student academic criteria, family income brackets, and demographics against Tamil Nadu state and central scholarship guidelines.
- Built an interactive web dashboard providing eligibility qualification scoring, required document verification checklists, and application deadline alerts.

Smart Precision Fertilization System — IoT Sensor-Based Agriculture **Barnstromz 2K26 Expo, 2026**

Technologies: Python, IoT Hardware (ESP32), Machine Learning, FastAPI

- Designed a sensor-based agriculture telemetry system presented at the Barnstromz 2K26 project expo (11 September 2026).
- Ingested real-time analog and digital telemetry streams from soil NPK, moisture, and pH sensors using ESP32 microcontrollers.
- Evaluated soil conditions against optimal nutrient curves to recommend targeted bio-fertilizer dosages and mitigate environmental chemical runoff.

ADDITIONAL TECHNICAL PROJECTS

Java Banking Management System & Ledger Engine **GitHub Repository, 2024**

Technologies: Java (OOP), File Persistence, Input Validation, Collections Framework

- Implemented a console banking transaction application demonstrating strict Object-Oriented Principles (encapsulation, abstraction, inheritance, polymorphism).
- Built account creation, PIN verification, deposit, withdrawal, and transaction logging routines with balance boundary checks to prevent arithmetic underflows.

Cognifyz Full-Stack Engineering Suite **GitHub Repository, 2024**

Technologies: Node.js, Express.js, REST APIs, JavaScript, Bootstrap, EJS

- Completed 5 end-to-end full-stack web application modules with Node.js and Express REST endpoints during internship track.
- Implemented structured payload validation, clean middleware routing, and dynamic client-side interface rendering.

WORK EXPERIENCE & INTERNSHIPS

Digital Marketing & Web Operations Intern, Skypark ITech **11 – 27 August 2025**

- Managed digital marketing initiatives, search engine indexing workflows, and content optimization strategies for web assets.
- Analyzed user engagement metrics and improved online visibility across targeted regional search demographics.

Full-Stack Development Intern, Cognifyz Technologies **Remote, 2024**

- Engineered responsive client-side web components and integrated backend RESTful endpoints with Node.js and Express.
- Authored clean, maintainable code following standardized software design principles across 5 production deliverables.

Founder & Lead Technology Developer, PosterKadai (Custom Art E-Commerce) **Tamil Nadu, India, 2025 – Present**

- Developed and deployed the entire digital product infrastructure on Google Firebase, managing catalog indexing and direct WhatsApp order channels.

Program Organizer & Technical Curriculum Lead, Image Color Lab **Online, 2025 – 2026**

- Organized a 15-day technical design and digital assets internship program, mentoring students on digital layouts and design fundamentals.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering **Third-Year (2024 – 2028)**

Dhanalakshmi Srinivasan College of Engineering, Coimbatore, Tamil Nadu, India

- Affiliated with Anna University. Core Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java/C++), Database Management Systems (SQL & NoSQL), Operating Systems, Computer Networks, Software Engineering.
- Actively building and deploying production software and participating in collegiate technical exhibitions alongside academic curriculum.

HACKATHONS, EXHIBITIONS & CERTIFICATIONS

- **Smartathon '26 (DSU × ONEYES Hackathon):** Participant (9–10 September 2026) — Built *TN Scheme AI* government scholarship and welfare discovery platform.
- **Barnstromz 2K26 Project Expo:** Exhibitor (11 September 2026) — Demonstrated the *Smart Precision Fertilization System* integrating IoT soil sensors and predictive analysis.
- **MongoDB Tech-Odyssey:** Developed full-stack agricultural direct trade platform *KisanConnect* utilizing MongoDB Atlas atomic operators.
- **IIT Guwahati AI & ML Coursework Qualification:** Qualified and completed machine learning foundations under IIT Guwahati E&ICT Academy.
- **Full-Stack Development Certification:** Verified internship completion certificate from Cognifyz Technologies.
- **Open-Source Code Contributions:** 450+ verified Git commits across 9 active technical repositories on GitHub (github.com/Surya252006).